

Blender 2.0

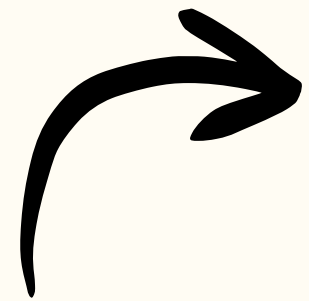
Presented by Hailie Tucker



Date	GBM	GBM Title	Secondary Event	Presenter
9 Jan		(No Meeting)	(Involvement fair?)	
15 Jan	1	Intros, Icebreakers, and Retrospectives		Everyone
21 Jan	2	Unity 201 (what new things?)	In the Co-Lab	Jake
29 Jan	3	2D Art -- Pixel and ...not pixel?	In the Pitch Room ↓	Calvin & Ares
5 Feb	4	2D Animation -- Frames and Tweening	(May have to move)	Calvin & Ares
12 Feb	5	Making Music for Non-Musicians		Marsh
19 Feb	6	Blender 201 -- Tidbits		Hailie
26 Feb	7	Unity 202 -- Lighting	Florida Comedy Jam (27-1)	Jake & Marsh
5 Mar	8	Writing For Games		Screenwriting Club
12 Mar	9	1 HOUR LEVEL DESIGN JAM		David
19 Mar		Spring Break		🦋
26 Mar	10	Sound Design and the Art of Foley	(May have to move)	Yusong
2 Apr	11	Unity 203 -- A*, Agents and Navmesh		Jake & Marsh
9 Apr	12	Online Portfolios (HTML Basics) -WICS collab		Hailie
16 Apr	13	Prototyping Levels	(May have to move)	Marsh
23 Apr	14	Year's Out Celebration!		Everyone

we are here



 <https://itch.io/jam/florida-comedy-game-jam-26>



DEVLUP @ FSU X UF X FIT X USF X UCF

Make the funniest game in Florida.

Florida Comedy Game Jam

February 27th 6:00 PM - March 1st 6:00PM
at the Garnet Gaming Lounge

A promotional poster for the Florida Comedy Game Jam. The background is a dark stage with red curtains. In the center, a 3D model of a clown with a red nose, a red and yellow polka-dot jacket, and blue and green polka-dot pants stands with arms outstretched. A black microphone on a stand is positioned to the left of the clown. Overlaid on the image are two windows from the Unity game engine: a 'Hierarchy' window on the left showing a tree structure with 'MainMenu Canvas' selected, and an 'Inspector' window on the right showing the properties of the 'MainMenu Canvas' component, including its transform, anchors, and rendering settings. In the bottom left corner, there is a QR code. In the bottom right corner, there are three logos: the Florida State University (FSU) logo, the Student Gaming Association (SGA) logo, and the Garnet Gaming Lounge logo, which includes the text 'HOME OF FSU ESPORTS'.

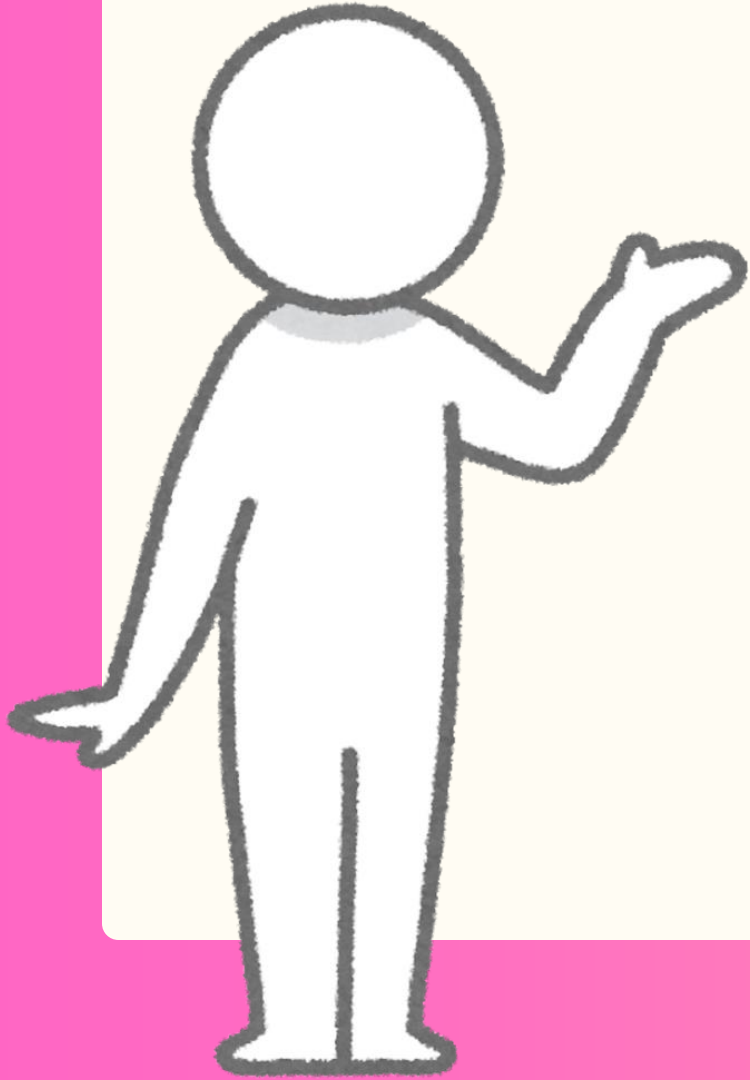
Showcase!



Blender Interface Overview

Understanding Your 3D Workspace - Movement

- **Rotate View:** Click and hold the Middle Mouse Button (Mouse-M).
- **Translate (Pan) View:** Shift + Mouse-M.
- **Zoom View:** Mouse-Wheel for stepwise zooming, or Ctrl + Mouse-M for continuous zooming.
- **Focus on Selected:** Numpad . (period) to center the view on the active object.
- **Toggle Camera View:** Numpad 0.
- **View Pie Menu:** ` (backtick) opens a menu to quickly switch between Top, Front, Right, and other views



Modes

Object Mode

- Add new objects
- move object from origin
- scale object
- measure



SHIFT + SPACEBAR

Edit Mode

- Vertex, Edge, and Faces
- Extrude (E)
- Inset (I)
- Bevel (ctrl + B)
- Loop cut (ctrl + R)
- Knife (K) -->to escape hit (enter)
- Spin (shift + 0)
- smooth (ctrl + 1)

Scrolling can effect these!

Special keys

Best use is to simplify complex models and get rid of unnecessary overlapping vertices

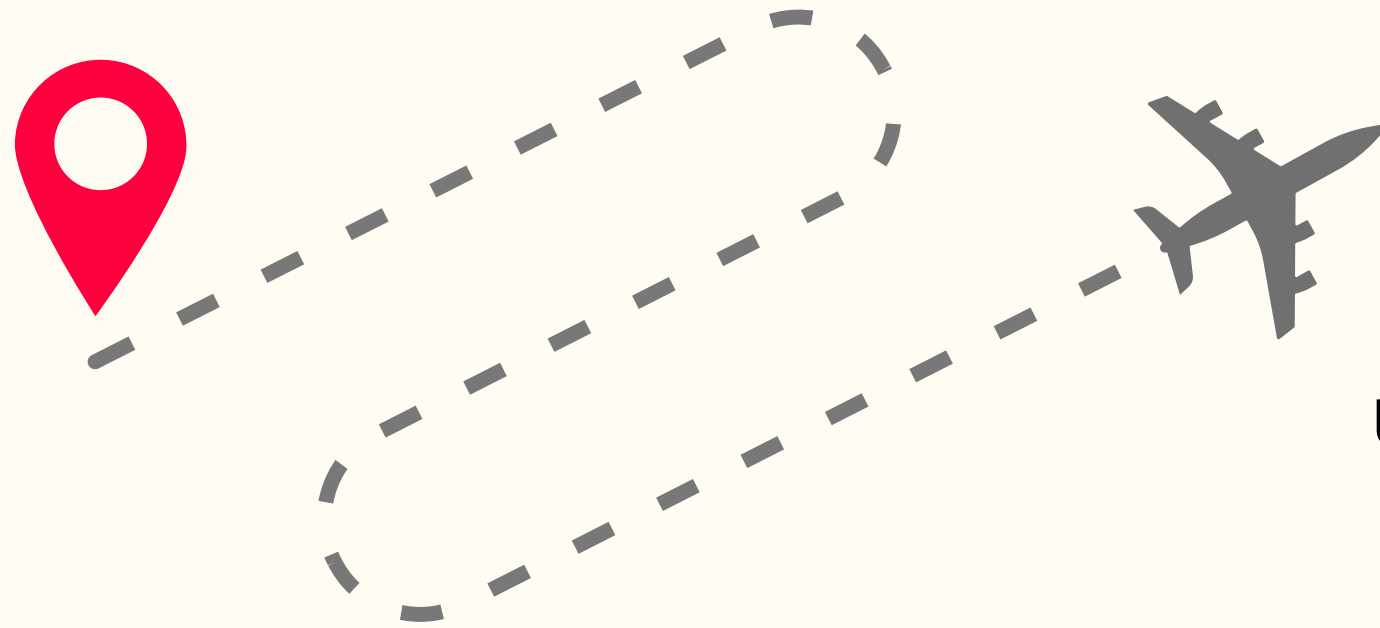
Key	Outcome
a	selects all
m	merge
d	by distance

Ex. When importing .svg files (text)

- F3 (Search): If you forget where a tool is, press F3 and type its name.



Special keys



ctrl + right click

places vertices!

Use: to trace an image

step 1: drop image in blender

step 2: merge object to 1 vertex

step 3: use the x,y,z sphere to align camera

**step 4: use ctrl + right click to add vertices along
the edges of the object you wish to trace**



More special keys

f	creates edges or faces
ctrl + right click	when extruding, extrudes to the cursor
cmd + a or control + a (on window)	opens apply window
right click + set origin	change where origin is
s	scale



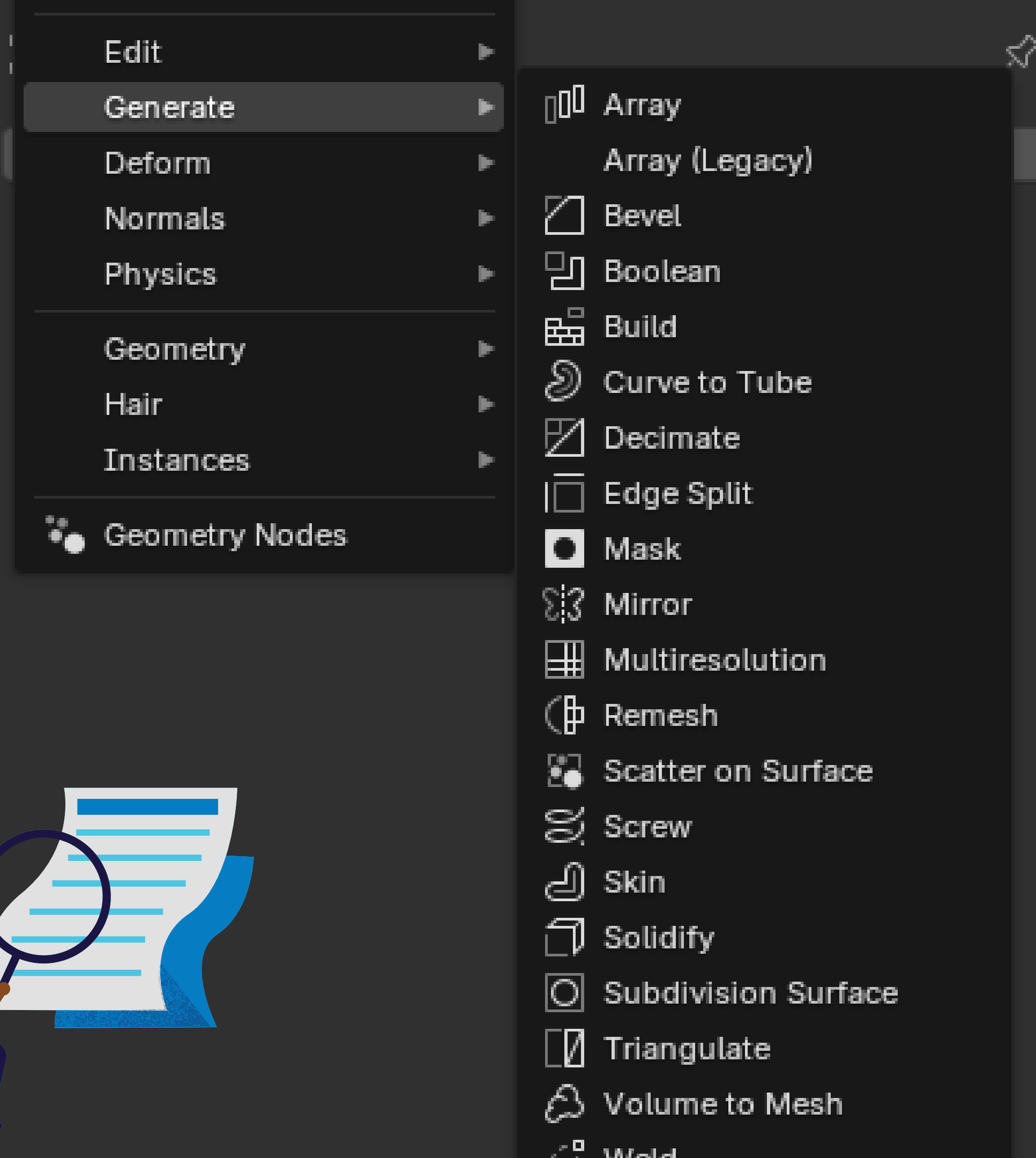
Modifiers

Generate

There are 21 generate modifiers
so, we shall NOT go over them all today

- Array
- Boolean
- Mirror
- Screw
- Solidify
- Subdivision Surface

(if multiple, order matters, reads top down)

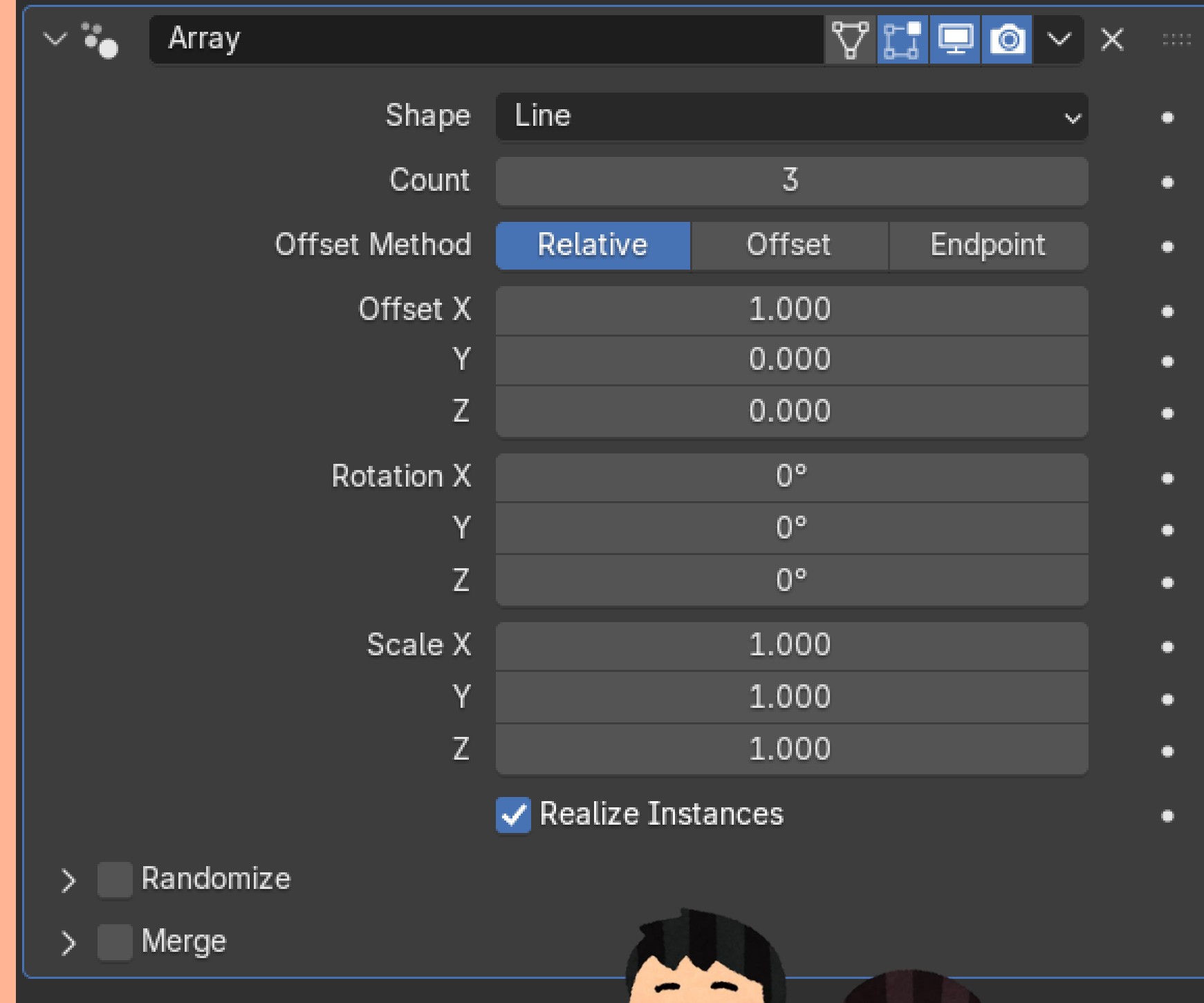


Modifiers

Array

Creates multiple repeating copies of your object in a sequence. It is the go-to tool for generating repetitive structures like chains, fences, brick walls, or stairs .

- **Relative / Constant Offset:** Spaces the copies out in a straight line based either on the object's own size or by exact measurements.
- **Object Offset:** Uses a secondary object (like an "Empty") to control the movement, rotation, and scale of the copies. This is how you create complex circular, curving, or spiraling patterns.

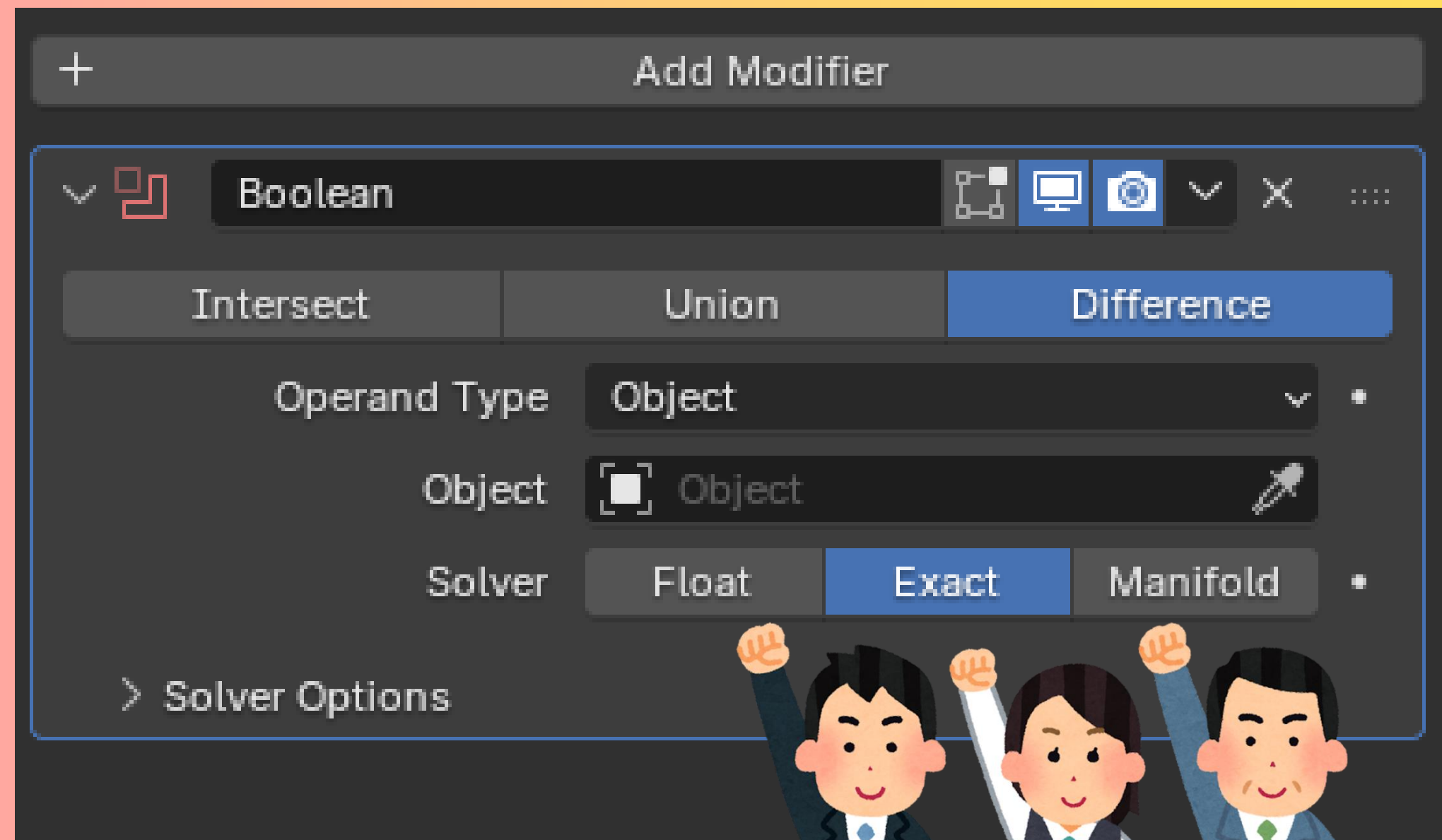
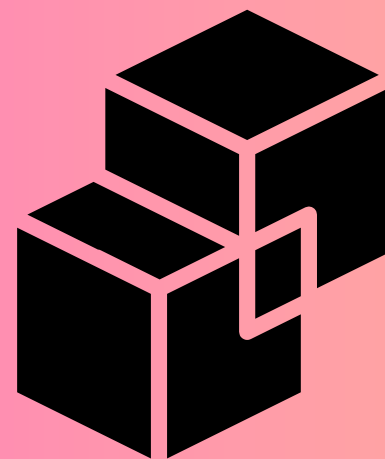


Modifiers

Boolean

Performs logical operations using another object to reshape your mesh.

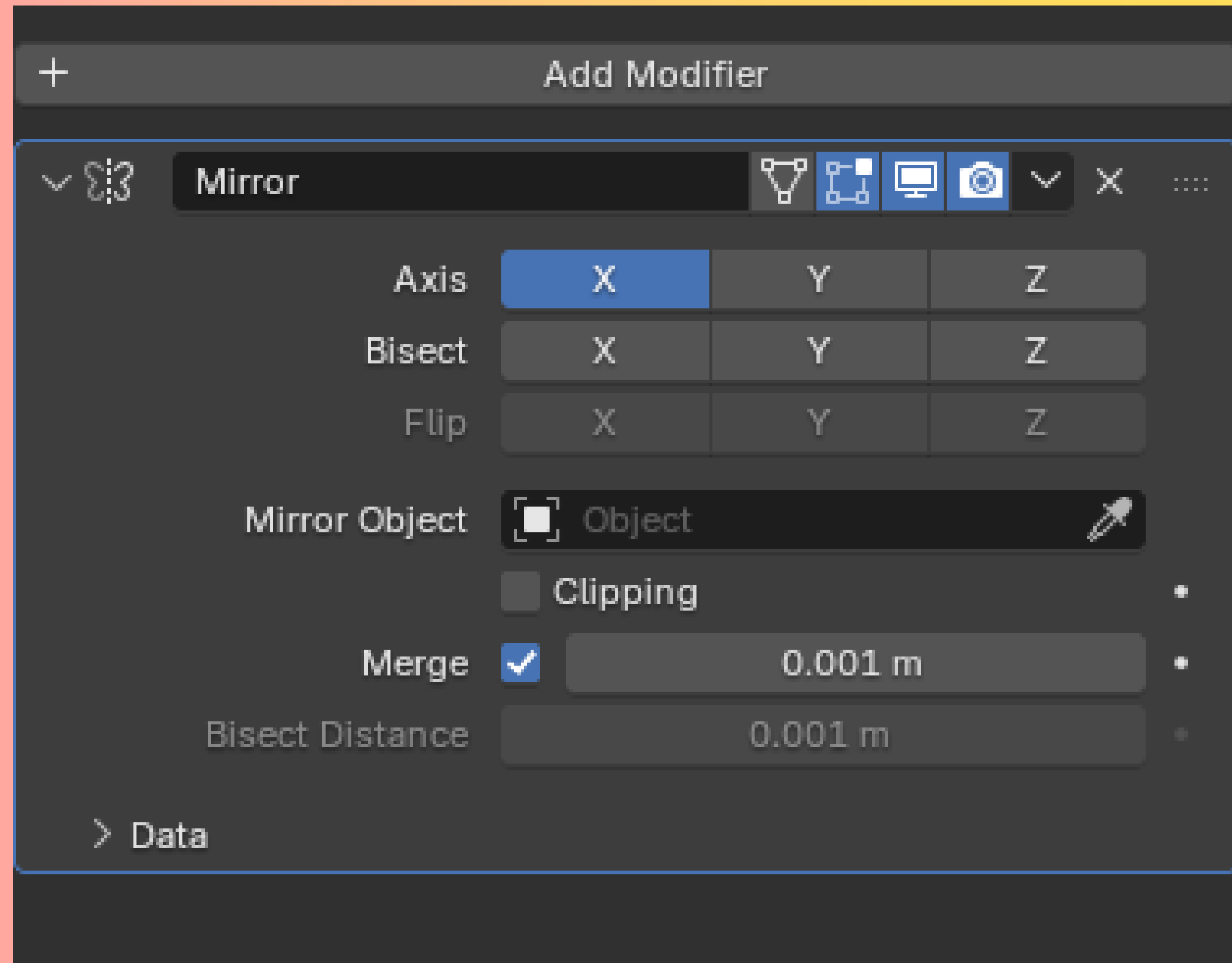
- **Difference:** Carves one shape out of another (like a hole punch).
- **Union:** Fuses two shapes together into a single mesh.
- **Intersect:** Keeps only the volume where the two shapes overlap.



Modifiers

Mirror

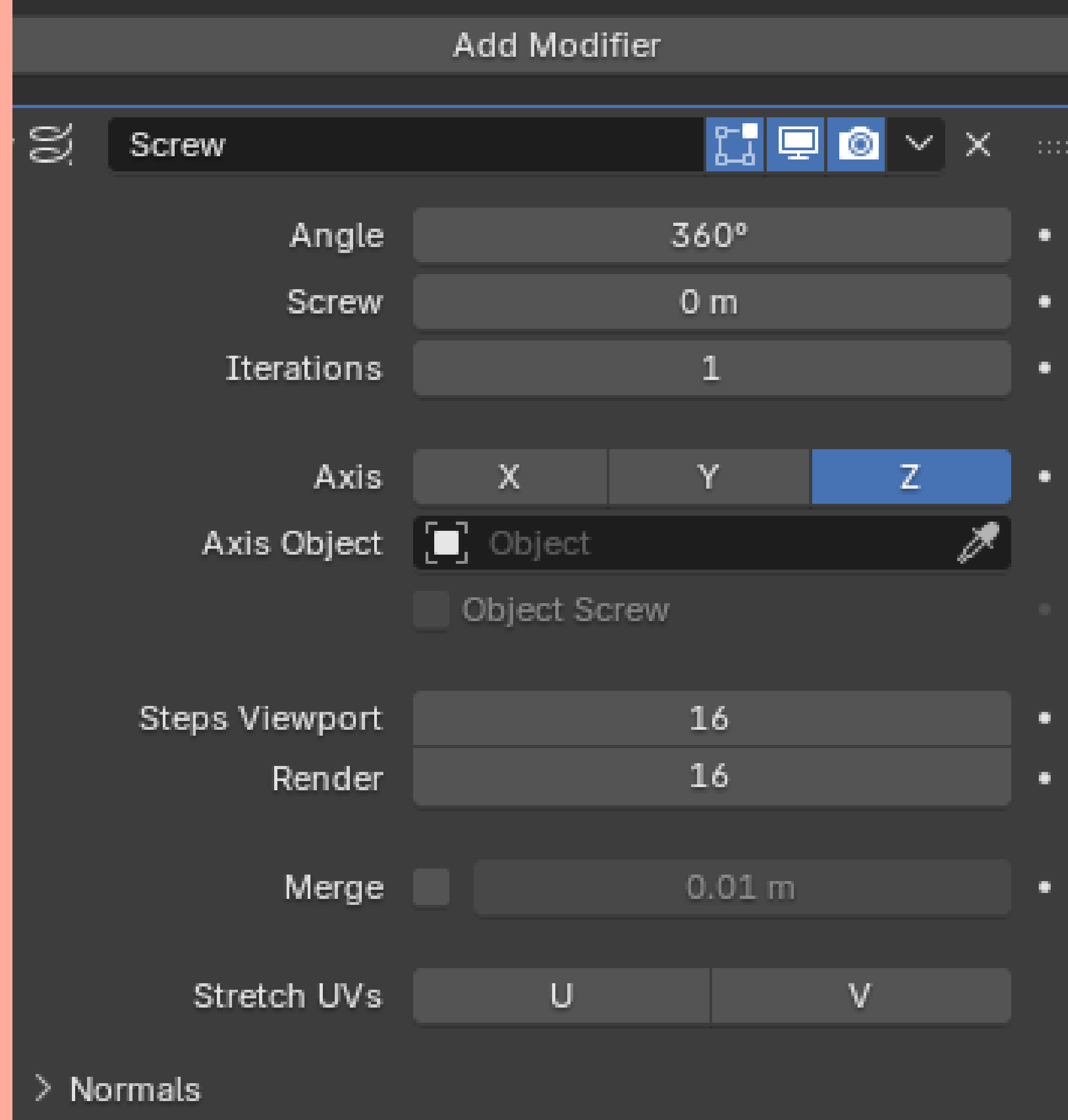
Symmetrically duplicates your geometry across a chosen axis (X, Y, or Z). This is essential for modeling symmetrical objects like faces, cars, or aircraft, allowing you to work on just one half while the other updates automatically



Modifiers

Screw

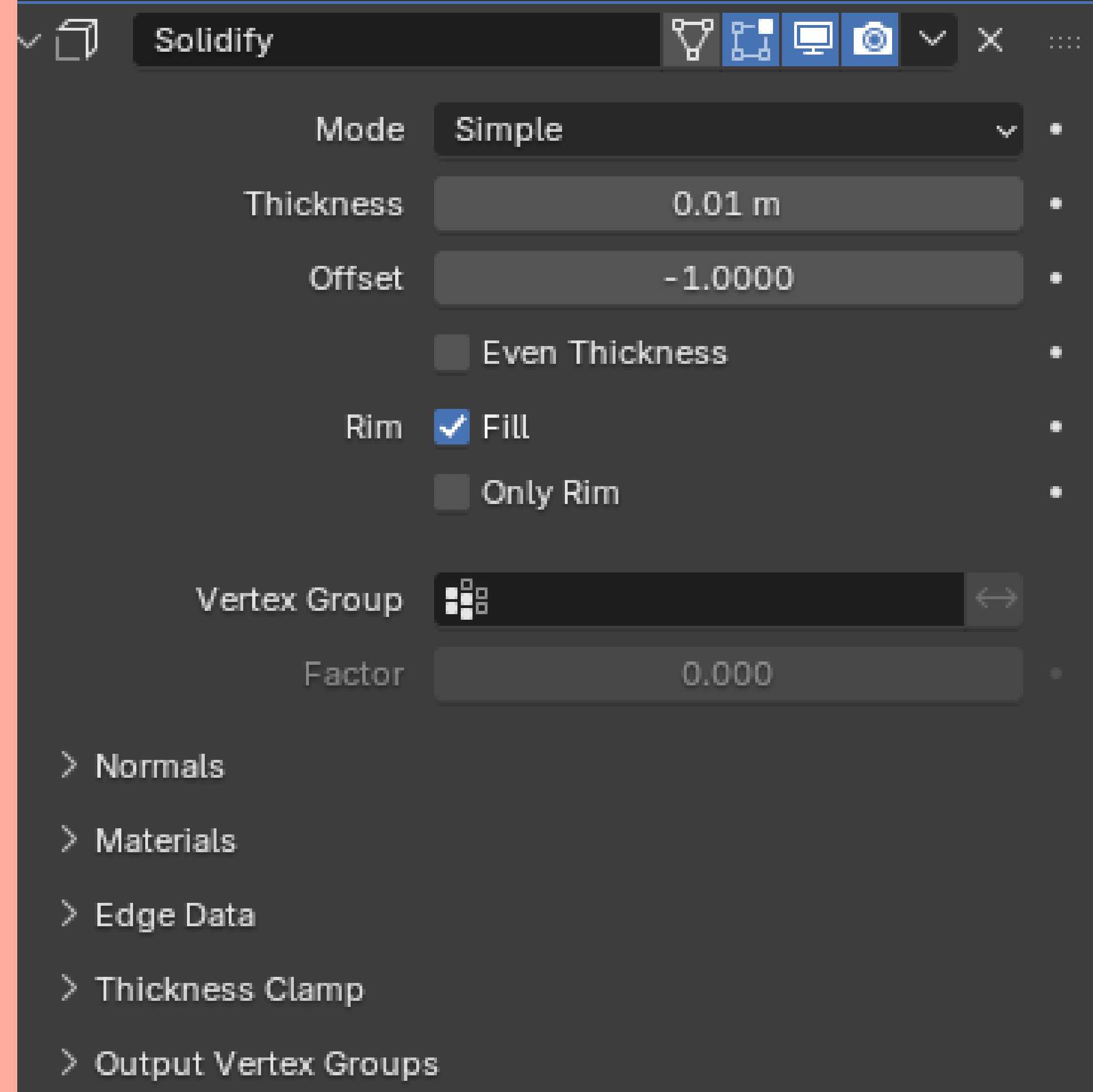
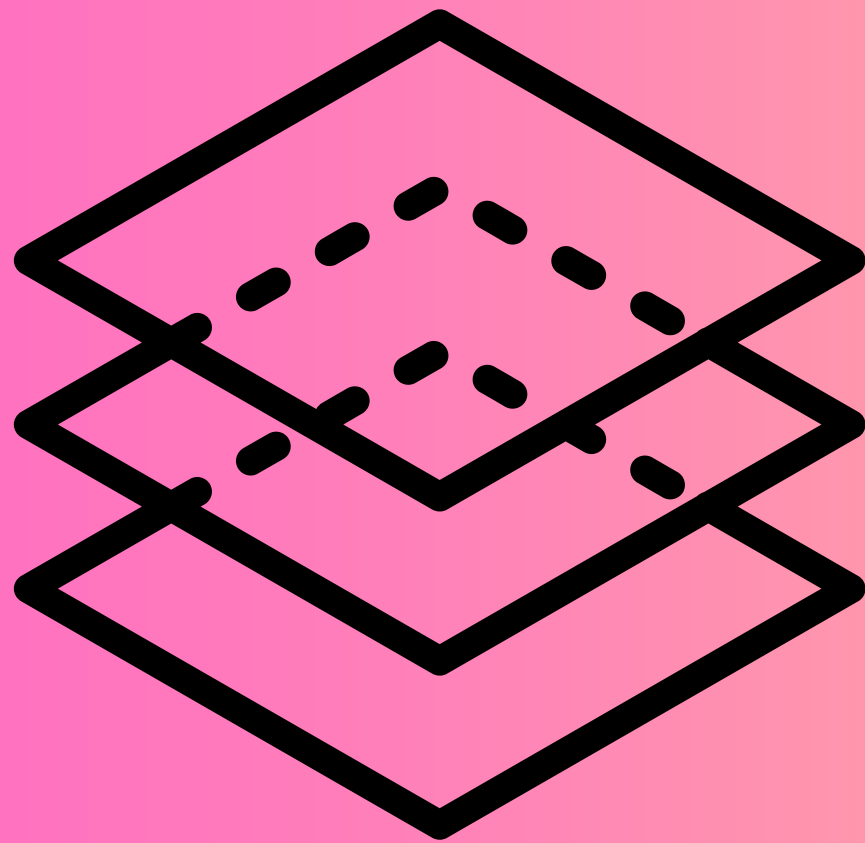
Takes a 2D profile (like a line or curve) and spins it around an axis to create a 3D spiral or helix. It is used to generate screws, springs, bolts, and spiral staircases.



Modifiers

Solidify

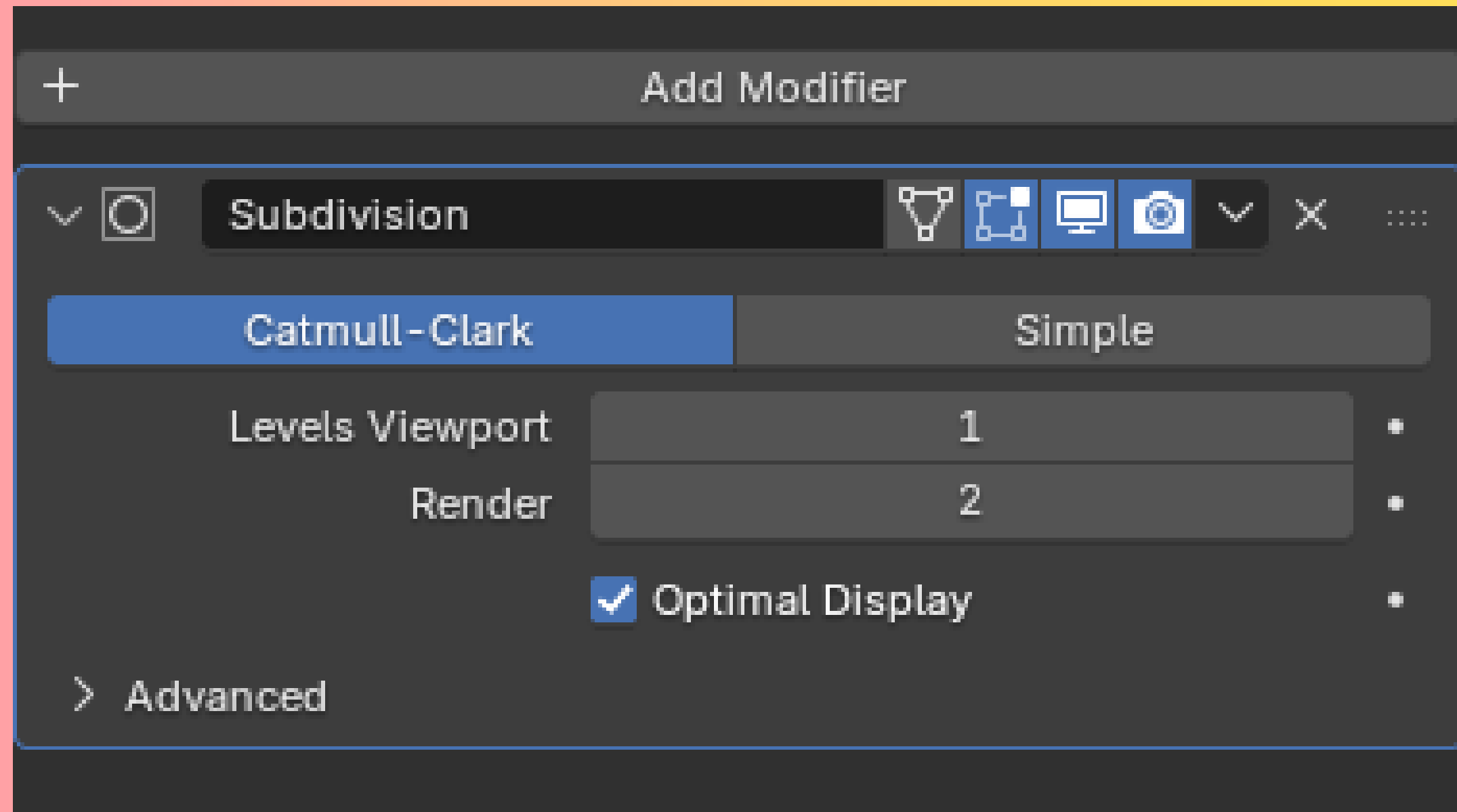
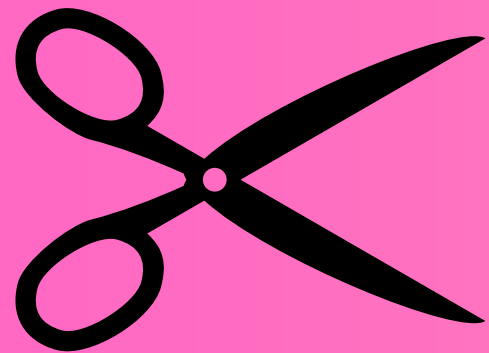
Adds depth or thickness to a flat, paper-thin surface. It instantly turns a 2D plane into a 3D object with volume, making it perfect for walls, armor plating, or cloth.



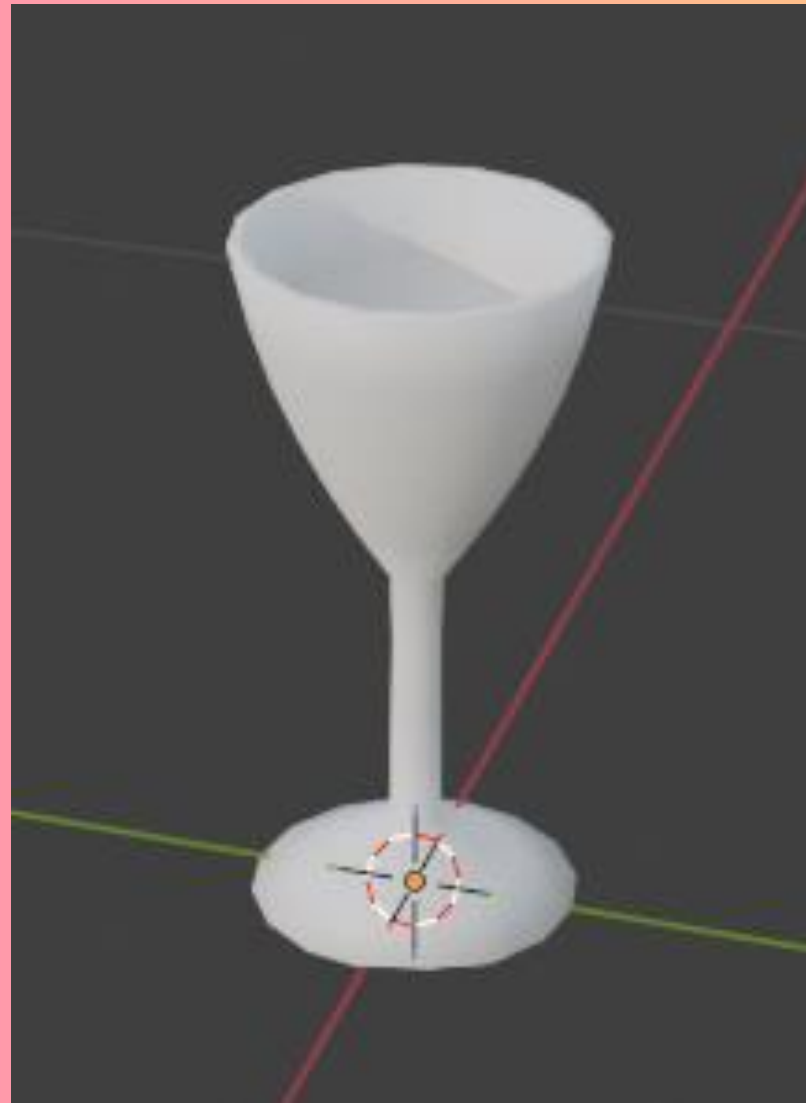
Modifiers

Subdivision Surface

Divides existing faces into smaller ones to smooth out the geometry. It is the standard method for taking a low-poly model and turning it into a smooth, high-resolution model.



Lets Test what you learned!



Model a cup/glass

Put your projects in
showoff!



Open for Questions/ Research



Hopefully y'all learned something!

